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# Priority Scheduling Program (Non-Preemptive)

n = int(input("Enter number of processes: "))

processes = []

# Input process details
for i in range(n):
    at = int(input(f"Enter arrival time for process {i + 1}: "))
    bt = int(input(f"Enter burst time for process {i + 1}: "))
    pr = int(input(f"Enter priority for process {i + 1} (smaller number = higher priority): "))

    processes.append({
        "pid": i + 1,
        "arrival": at,
        "burst": bt,
        "priority": pr,
        "completed": False,
        "waiting": 0,
        "turnaround": 0
    })

time = 0
completed = 0

while completed < n:
    # Get processes that have arrived and are not completed
    ready_queue = [p for p in processes if p["arrival"] <= time and not p["completed"]]

    if ready_queue:
        # Select process with highest priority (lowest number)

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50     if ready_queue:
51         # Select process with highest priority (lowest number)
52         current = min(ready_queue, key=lambda x: x["priority"])
53
54         # Calculate waiting time
55         current["waiting"] = time - current["arrival"]
56
57         # Execute process
58         time += current["burst"]
59
60         # Calculate turnaround time
61         current["turnaround"] = current["waiting"] + current["burst"]
62
63         current["completed"] = True
64         completed += 1
65     else:
66         # If no process is ready, move time forward
67         time += 1
68
69 # Display results
70 print("\nPID\tAT\tBT\tPR\tWT\tTAT")
71 for p in processes:
72     print(f"{p['pid']}\t{p['arrival']}\t{p['burst']}\t{p['priority']}\t{p['waiting']}\t{p['turnaround']}")
73
74 # Calculate averages
75 avg_wt = sum(p["waiting"] for p in processes) / n
76 avg_tat = sum(p["turnaround"] for p in processes) / n
77
78 print(f"\nAverage Waiting Time: {avg_wt:.2f}")
79 print(f"\nAverage Turnaround Time: {avg_tat:.2f}")

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Enter number of processes: 4
Enter arrival time for process 1: 9
Enter burst time for process 1: 8
Enter priority for process 1 (smaller number = higher priority): 7
Enter arrival time for process 2: 8
Enter burst time for process 2: 6
Enter priority for process 2 (smaller number = higher priority): 9
Enter arrival time for process 3: 7
Enter burst time for process 3: 5
Enter priority for process 3 (smaller number = higher priority): 9
Enter arrival time for process 4: 5
Enter burst time for process 4: 6
Enter priority for process 4 (smaller number = higher priority): 8
```

PID	AT	BT	PR	WT	TAT
1	9	8	7	2	10
2	8	6	9	11	17
3	7	5	9	18	23
4	5	6	8	0	6

Average Waiting Time: 7.75

Average Turnaround Time: 14.00